Making the knights

- The turning centres and the hole through the knight N1 should be marked out on a square or rectangular cross-sectioned blank. The hole is then drilled. Each workpiece is then roughed between centres, chucked axially in the pin chuck, finish-turned, and later polished in the lathe and parted off.
- Knight N2 is turned on two axes B-B and C-C. The workpiece should be at least 75mm long and have a cross section of at least 45 x 45mm.

1) Accurately mark out and punch the two axes on the ends of the workpieces.
2) Mount a workpiece between centres on axis B-B and accurately turn it to a cylinder of 42mm diameter.
3) Mark out, finish-turn and sand the 10mm wide x 3mm deep channel in the outside of the cylinder. If you allow, say, 10mm of waste at the right-hand end of the man, the right-hand side of the channel should be 21mm in from the right-hand end of the workpiece.
4) Remount the workpiece between centres on axis C-C. Finish-turn and sand the cylindrical body of N2. Turn a suitable chucking spigot at the workpiece's left-hand end.
5) Remount the workpiece on axis C-C using the chucking spigot. Use a series of arcing cuts with a skew's long point to cut the top of N2. Partially cut the bottom of N2. Sand.
6) The N2 men can be later remounted for polishing using the chucking spigot, and be completely parted off when the polishing is finished.

- To make the knights N3, you turn a ring doughnut form for each pair of knights – because the grain runs vertically through these knights, only two N3 knights can be sawn from each doughnut. Apart from the difference between the ring workpiece cross sections, the procedure is identical to that described for the knight in article three except that the backing plate usually can't be reused. The steps are:

1) Prepare a disc-shaped workpiece, mount it on a commercial screw chuck, and finish-turn and sand its right-hand half.
2) On the same screw chuck turn a backing plate with a shallow annular recess deepest at 46mm diameter. On the opposite side of the backing plate mark out the saw cuts that will later be made to separate the two knights N3.
3) Screw the already finish-turned and sanded side of the doughnut into the recess in the backing plate.
4) Rechuck the backing plate and doughnut assemblage on the screw chuck.

Drilling the 16mm diameter hole in a knight N1. Note that the turning centres have also been punched

Parting off a black knight N1 after polishing

Checking the cross section of the right-hand half of a knight N3 doughnut with a template cut from a plastic milk container

Left to right: a commercial screw chuck, a backing plate and a half-turned doughnut

5) Finish-turn and sand the right-hand side of the doughnut.
6) Bandsaw the two knights free from the doughnut.

- The cranked knights N4 and N5 are turned in one piece between centres on two parallel axes D-D and E-E. Knight N4 has a horizontal shoulder; alternative knight N5 has a sloping shoulder instead. The steps in turning N4 or N5 are:

1) Mark out and punch the centres D and E in each end of a workpiece.
2) Mount the workpiece between centres on axis E-E. Rough the workpiece to a cylinder of about 42mm diameter.
3) Leaving about 10mm of waste at the right-hand end of the workpiece, cut the junction between the upper and power cylinders with a skew's long point. Use small gentle cuts so that the skew's long point isn't forced off line. Finish-turn the periphery of the upper (right-hand) cylinder. Also turn a cylindrical chucking section at the workpiece's left-hand end.
4) Remount the workpiece between centres on axis D-D. Finish-turn and sand the lower cylinder of the knight.
5) Chuck the workpiece by the pin on its left-hand end on axis E-E. Turn off the right-hand waste and sand the upper cylinder.

- Knights N6 are turned in two pieces. A pin turned on the top section is later glued into a suitably positioned hole bored into the top of the bottom section. However, making knight N6 is considerably more fiddly than turning N4 or N5. It's therefore worth practising making N4 or N5, and thus avoiding making N6.

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